

Determinants I HW solutions

$$1) A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, B = \begin{bmatrix} x & y \\ z & w \end{bmatrix}, \det(A) = ad - bc \\ \det(B) = xw - yz$$

$$AB = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x & y \\ z & w \end{bmatrix} = \begin{bmatrix} ax+bz & ay+bw \\ cx+dz & cy+dw \end{bmatrix}$$

$$\begin{aligned} \det(AB) &= (ax+bz)(cy+dw) - (ay+bw)(cx+dz) \\ &= axcy + adxw + bzcw + bzdw \\ &\quad - ayx - aydz - bwcx - bydz \\ &= adxw + bzcw - aydz - bwcx \\ &= ad(xw - yz) - bc(xw - yz) \\ &= (ad - bc)(xw - yz) \\ &= \det(A)\det(B) \quad \checkmark \end{aligned}$$

$$2) A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, k \text{ an arbitrary scalar}$$

$$\det(A) = ad - bc, \text{ so } k\det(A) = kad - kdc$$

$$\begin{aligned} \det(kA) &= \begin{vmatrix} ka & kb \\ kc & kd \end{vmatrix} = kakd - kbkc \\ &= k^2ad - k^2bc \\ &= k^2(ad - bc) \end{aligned}$$

$$\det(kA) \neq k\det(A), \boxed{\det(kA) = k^2\det(A)}$$

$$3) \det(A) = ad - bc, \det(B) = bc - ad, \det(C) = bc - ad$$

a) $\det(A) = -\det(B)$, they are each other's additive inverse.

$$b) \det(A) = -\det(C)$$

4) Answers vary here for (a) and (b), but for

(c)(a) and (c)(b) you should see that the determinant is negated whenever you swap 2 rows or columns.

(c)(c) You should see that the determinant increases by a factor of k .

(c)(d) The determinant increases by a factor of k^3 .